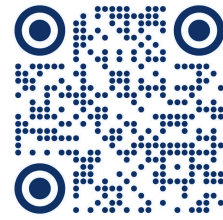


CURRICULUM VITAE

Prof. Olivier-Gilles MARIN, PhD



Residence: Shanghai, China
Permanent Residence Permit

Mobile: +86-185 1218 5704
E-Mail: racine75@gmail.com
<https://olivier-marin.github.io/>

Profile Summary

I am a computer scientist working at the intersection of distributed systems, collective artificial intelligence, and human-machine systems, with extensive experience in academic leadership and interdisciplinary program development. My research examines how large-scale intelligent systems coordinate, adapt, and remain interpretable when operating in real-world environments involving both machines and human decision-makers. This work connects theoretical models of distributed coordination with applied system design in domains such as infrastructure, networks, and quantum computing.

Alongside my research, I have led the creation and development of academic programs and research initiatives across Europe and China. I have served in senior leadership roles including Associate Dean and Interim Dean, with responsibility for curriculum architecture, faculty development, and the integration of engineering research with design-driven, application-oriented education. Building on this experience, my interests increasingly center on applied graduate education and research structures that bridge intelligent systems, design practice, and industry-oriented innovation in urban settings.

Academic appointments

- **Chair Professor, Shanghai University of Engineering Science, China** – 2026 - present
 - **Head of the Humans+Machines graduate program** – 2026 - present
- **Maître de Conférences (*tenured - on hold*), Sorbonne Université, France** – 2004 - present
- **Professor of Practice in Computer Science, NYU Shanghai, China** – 2018 - 2026
 - **Associate Dean of Arts and Sciences, NYU Shanghai** – 2022 - 2024
 - **Interim Dean of Arts and Sciences, NYU Shanghai** – 2021 - 2022
- **Associate Professor of Practice in Computer Science, NYU Shanghai** – 2015 - 2018
- **Postdoctoral Researcher, Vrije Universiteit, Amsterdam** – 2003 - 2004

Academic Leadership and Program Development

- Head of Computer Science, NYU Shanghai
- Interim Dean and Associate Dean, Arts and Sciences, NYU Shanghai
- Co-founder and Co-director of an industry-oriented BSc program, Sorbonne Université
- Vice-President of the Faculty Recruitment Committee, Sorbonne Université

Funded Research

ANR PRC – FrugalDiNet (2024–2027)

Co-PI, collaboration with three French institutions. This project develops frugal distributed algorithms at the network layer using intelligent hardware such as programmable switches and Data Processing Units (DPUs). My contribution focuses on leveraging low-level telemetry data to design better datacenter services, enhancing resilience and scalability with minimal overhead.

STCSM – Scalable Collective AI (2025–2028)

Collaboration with Mathieu Lauriere, NYU Shanghai. This project studies coordination and learning in large populations of agents using mean-field models and reinforcement learning. I lead the distributed simulation and implementation, building scalable infrastructures to validate theoretical results in applied settings.

Selected former international collaborations

Inria/CONICYT Associate Team ARMADA (Chile), PI (2014–2017); *PHC MAIMONIDE* (France/Israel), PI (2014–2016); *Multiple doctoral grants* (CONICYT, Inria, French Ministry of Higher Ed.); *Eurocontrol FTATC* (Fault-Tolerant Air Traffic Management), Vice-Coordinator (2006–2007).

Graduate Supervision

12 MSc theses in distributed systems

3 PhD students defended:

Rudyard Cortès (2017): **Scalable location-temporal range query processing**

Maxime Véron (2015): **Scalable services for massively multiplayer online games**

Erika Rosas (2012): **Building trustworthy services in peer-to-peer networks**

International Publications

Journal articles

1. P. Sens, L. Arantes, A. G. De Moraes Rossetto, O. Marin (2024) *Stab-FD: a Cooperative and Adaptive Failure Detector for Wide Area Networks*, Journal of Parallel and Distributed Computing (JPDC) 186.
2. X Bonnaire, R Cortes, F Kordon, O Marin (2017) *ASCENT: a Provably-Terminating Decentralized Logging Service*, The Computer Journal 60: 12. 1889–1911
3. Erika Rosas, Nicolas Hidalgo, Veronica Gil-Costa, Carolina Bonacic, Hermes Senger, Mauricio Marin, Luciana Arantes, Cesar Marcondes, Olivier Marin (2016) *Survey on Simulation for Mobile Ad-hoc Communication for Disaster Scenarios*, Journal of Computer Science and Technology.
4. Rudyar Cortés, Xavier Bonnaire, Olivier Marin, Pierre Sens (2015) *Stream Processing of Healthcare Sensor Data : Studying User Traces to Identify Challenges from a Big Data Perspective* Procedia Computer Science 52: 1004-1009.
5. Maxime Véron, Olivier Marin, Sébastien Monnet, Zahia Guessoum (2014) *Towards a scalable refereeing system for online gaming* Multimedia Systems 20: 5. 579-593
6. Erika Rosas, Olivier Marin, Xavier Bonnaire (2011) *CORPS : Building a Community Of Reputable PeerS in Distributed Hash Tables* Computer Journal 54: 10. 1721-1735
7. Zahia Guessoum, Jean-Pierre Briot, Nora Faci, Olivier Marin (2010) *Towards reliable multi-agent systems : An adaptive replication mechanism* Multiagent and Grid Systems 6: 1.

Book chapters

1. Olivier Marin, Gaël Thomas, Sébastien Monnet (2011) *Peer-to-Peer storage In: Distributed Systems : Design and Algorithms* Edited by: Serge Haddad, Fabrice Kordon, Laurent Pautet, Laure Petrucci. 59-80 John Wiley & Sons, Ltd.
2. Olivier Marin, Jean-Michel Busca (2008) *ePOST, une expérience de courrier électronique pair-à-pair* In: Systèmes répartis en action : de l'embarqué aux systèmes à large échelle Edited by: Fabrice Kordon, Laurent Pautet, Laure Petrucci. 257-282 Hermes.
3. Olivier Marin, Marin Bertier, Pierre Sens, Zahia Guessoum, Jean-Pierre Briot (2007) *DARX - A Self-Healing Framework for Agents In: Reliable Systems on Unreliable Networked Platforms*, Revised Selected Papers of the 12th Monterey Workshop Edited by: Fabrice Kordon, Janos Sztipanovits. 88-105 Springer-Verlag.
4. Zahia Guessoum, Jean-Pierre Briot, Olivier Marin, Athmane Hamel, Pierre Sens (2003) *Dynamic and Adaptive Replication for Large-Scale Reliable Multi-Agent Systems* In: Software Engineering for Large-Scale Multi-Agent Systems Edited by: Alessandro Garcia, Carlos Lucena, Franco Zambonelli, Andrea Omicini, Jaelson Castro. 182-198 Springer.

Conference papers

1. Xiaonan Li, Olivier Marin (2022) *Towards Implementing ML-Based Failure Detectors* In: EDCC'22: 18th European Dependable Computing Conference (Student Forum).
2. Hongyi Li, Hongyi Liu, Olivier Marin (2021) *Demystifying Distributed Systems with a Storytelling Method* In: SIGCSE '21: Proceedings of the 52nd ACM Technical Symposium on Computer Science Education (Poster presented at the ACM Student Research Competition).
3. Leyi Sun, Yifan Zhuo, Olivier Marin (2021) *Simple yet Efficient Deployment of Scientific Applications in the Cloud* In: 27th IEEE International Conference on Parallel and Distributed Systems (ICPADS 2021) Beijing, China.

4. Luciana Arantes, Roy Friedman, Olivier Marin, Pierre Sens (2017) *Probabilistic Byzantine Tolerance Scheduling in Hybrid Cloud Environments* In: 18th International Conference on Distributed Computing and Networking (ICDCN 2017) Hyderabad, India.
5. Florent Coriat, Anne Fladenmuller, Luciana Arantes, Olivier Marin (2016) *Crowdsourcing-based architecture for post-disaster Geolocation: a comparative performance evaluation* In: 15th IEEE International Symposium on Network Computing and Applications (NCA'2016).
6. Rudyar Cortés, Xavier Bonnaire, Olivier Marin, Luciana Arantes, Pierre Sens (2016) *GeoTrie: A Scalable Architecture for Location-Temporal Range Queries over Massive GeoTagged Data Sets* In: 15th IEEE International Symposium on Network Computing and Applications (NCA'2016).
7. Luciana Arantes, Marjorie Bournat, Roy Friedman, Olivier Marin, Pierre Sens (2015) *Elastic Management of Byzantine Faults* In: 11th European Dependable Computing Conference (EDCC-2015) (Fast Abstract).
8. Guillaume Turchini, Sebastien Monnet, Olivier Marin (2015) *Scalability and availability for massively multiplayer online games* In: 11th European Dependable Computing Conference (EDCC-2015) (Fast Abstract).
9. Erika Rosas, Felipe Garay, Nicolas Hidalgo, Olivier Marin (2015) *Mobility-aware DTN protocols for post-disaster scenarios* In: International Conference on Information and Communication Technologies for Disaster Management (ICT-DM'2015).
10. Maxime Véron, Olivier Marin, Sébastien Monnet, Pierre Sens (2015) *RepFD - Using reputation systems to detect failures in large dynamic networks* In: 44th International Conference on Parallel Processing (ICPP-2015).
11. Luciana Arantes, Roy Friedman, Olivier Marin, Pierre Sens (2015) *Probabilistic Byzantine Tolerance for Cloud Computing* In: 34th International Symposium on Reliable Distributed Systems (SRDS'15).
12. Rudyar Cortés, Xavier Bonnaire, Olivier Marin, Pierre Sens (2015) *FreeSplit : A Write-Ahead Protocol to Improve Latency in Distributed Prefix Tree Indexing Structures* In: 29th IEEE International Conference on Advanced Information Networking and Applications (AINA-2015) Gwangju, South Korea.
13. Rudyar Cortés, Olivier Marin, Xavier Bonnaire, Luciana Arantes, Pierre Sens (2015) *A Scalable Architecture for Spatio-Temporal Range Queries over Big Location Data* In: 14th IEEE International Symposium on Network Computing and Applications (NCA'15) Cambridge, MA, United States.
14. Florent Coriat, Luciana Arantes, Olivier Marin, Anne Fladenmuller, Nicolas Hidalgo, Erika Rosas (2014) *Towards distributed geolocation for large scale disaster management* In: 33rd International Conference of the Chilean Computer Science Society (SCCC 2014) Talca, Maule, Chile.
15. Maxime Véron, Olivier Marin, Sébastien Monnet (2014) *Matchmaking in multiplayer online games : studying user traces to improve the user experience* In: Proceedings of the 24th International Workshop on Network and OS Support for Digital A/V.
16. Fabrice Kordon, Xavier Bonnaire, Rudyar Cortés, Olivier Marin (2013) *Verification of a Quasi certification Protocol over a DHT* In: Dagstuhl Seminar "Formal Verification of Distributed Algorithms" Schloss Dagstuhl.
17. Xavier Bonnaire, Rudyar Cortés, Fabrice Kordon, Olivier Marin (2013) *A Scalable Architecture for Highly Reliable Certification* In: 12th IEEE International Conference on Trust, Security and Privacy in Computing and Communications – TrustCom 328-335 IEEE Computer Society.

18. Maxime Véron, Olivier Marin, Sébastien Monnet, Zahia Guessoum (2012) *Towards a scalable refereeing system for online gaming* In: 11th International Workshop on Network and Systems Support for Games (NetGames'2012) (Poster).
19. Adrien Debrie, Pacôme Eberhart, Pierre-Louis Roman, Cécile Le Pape, Olivier Marin (2012) *Towards a Geocentric Mobile Syndication System* In: IEEE International Conference on Internet of Things (iThings 2012).
20. Nicolas Hidalgo, Erika Rosas, Luciana Arantes, Olivier Marin, Pierre Sens, Xavier Bonnaire (2012) *Optimized range queries for large scale networks*. 26th IEEE International Conference on Advanced Information Networking and Applications (AINA-2012)
21. Nicolas Hidalgo, Erika Rosas, Luciana Arantes, Olivier Marin, Pierre Sens, Xavier Bonnaire (2011) *DRing : A Layered Scheme for Range Queries over DHTs* In: IEEE International Conference on Computer and Information Technology (CIT). 29-34.
22. Alysso N Bessani, Vinicius V Cogo, Miguel Correia, Pedro Costa, Marcelo Pasin, Fabrício A B Silva, Luciana Arantes, Olivier Marin, Pierre Sens, Julien Sopena (2010) *Making Hadoop MapReduce Byzantine Fault-Tolerant* In: Proceedings of the The 40th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN'2010) - Fast Abstract
23. Minh Nguyen-Duc, Zahia Guessoum, Olivier Marin, Jean-Francois Perrot, Jean-Pierre Briot, Vu Duong (2008) *Towards a Reliable Air Traffic Control* In: 7th International Conference on Autonomous Agents and Multiagent Systems (AAMAS'08) (Industrial Track) 101-104, ACM Press.
24. Minh Nguyen-Duc, Zahia Guessoum, Olivier Marin, Jean-Francois Perrot, Jean-Pierre Briot (2008) *A multi-agent approach to reliable air traffic control* In: 2nd International Symposium on Agent Based Modeling and Simulation (ABModSim'08).
25. Alessandro De De Almeida, Jean-Pierre Briot, Samir Aknine, Zahia Guessoum, Olivier Marin (2007) *Towards autonomic fault-tolerant multi-agent systems* In: 2nd Latin American Autonomic Computing Symposium (LAACS'07).
26. Xavier Bonnaire, Olivier Marin (2007) *Recursive Replication : A Survival Solution for Structured P2P Information Systems to Denial of Service Attack* In: 1st International Workshop on Peer to Peer Networks (PPN07) 931-940.
27. Jean-Pierre Briot, Zahia Guessoum, Samir Aknine, Alessandro De De Almeida, Nora Faci, Maïra Gatti, Carlos Lucena, Jacques Malenfant, Olivier Marin, Pierre Sens (2006) *Experience and Prospects for Various Control Strategies for Self-Replicating Multi-Agent Systems* In: Proceedings of the International Workshop on Software Adaptive and Self-Managing Systems (SEAMS'2006) 37-43 28th International Conference on Software Engineering (ICSE'2006), Shanghai, Chine.
28. Jean-Pierre Briot, Zahia Guessoum, Samir Aknine, Alessandro De De Almeida, Nora Faci, Jacques Malenfant, Olivier Marin, Pierre Sens (2006) *Engineering Control Strategies for Replication-Based Fault-Tolerant Multi-Agent Systems* In: International Workshop on Engineering of Fault-Tolerant Systems (EFTS'2006) 80-93 Luxembourg, Luxembourg
29. Samir Aknine, Olivier Marin (2005) *Role of Replication Planning for Fault Tolerant Multiagent Systems* In: Symposium on Adaptive Agents and Multi-Agent Systems (AAMAS).
30. David Mobach, Benno Overeinder, Olivier Marin, Frances Brazier (2005) *Lease-based Decentralized Resource Management in Open Multi-Agent Systems* In: 18th International Florida Artificial Intelligence Research Society Conference (FLAIRS '2005) 339-344.
31. Olivier Marin, Marin Bertier, Pierre Sens, Zahia Guessoum, Jean-Pierre Briot (2005) *DARX - A Self-Healing Framework for Agents* In: Proceedings of the 12th Monterey Workshop on Reliable Systems on Unreliable Networked Platforms Edited by:Fabrice Kordon, Janos Sztipanovits. 88-105 Laguna Beach, CA, USA

32. Marin Bertier, Olivier Marin, Pierre Sens (2003) *Performance Analysis of a Hierarchical Failure Detector* In: International Conference on Dependable Systems and Networks (DSN'03) IEEE Society Press.
33. Benno Overeinder, Frances Brazier, Olivier Marin (2003) *Fault tolerance in scalable agent support systems : integrating DARX in the AgentScape framework* In: 3rd IEEE/ACM International Symposium on Cluster Computing and the Grid 688-695.
34. Olivier Marin, Marin Bertier, Pierre Sens (2003) *DARX - A Framework for the Fault-Tolerant Support of Agent Software*. In: 14th. IEEE International Symposium on Software Reliability Engineering, (ISSRE 03).
35. Marin Bertier, Olivier Marin, Pierre Sens (2002) *Implementation and performance evaluation of an adaptable failure detector* In: International Conference on Dependable Systems and Networks (DSN) Edited by:IEEE Society Press. 354-363.
36. Jean-Pierre Briot, Zahia Guessoum, Sébastien Charpentier, Samir Aknine, Olivier Marin, Pierre Sens (2002) *Dynamic Adaptation of Replication Strategies for Reliable Agents* In: Proceedings of the Second Symposium on Adaptive Agents and Multi-Agent Systems (AAMAS-2) 13-21 AISB'02 Convention, London, UK: The Society for the Study of Artificial Intelligence and the Simulation of Behaviour (AISB).
37. Insaf Tnazefti-Kerkeni, Luciana Arantes, Olivier Marin (2002) *A Multi-Blackboard Approach to the Control/Monitoring of APS* In: WSEAS International Conference on Applied Mathematics and Computer Science (AMCOS'02).
38. Zahia Guessoum, Jean-Pierre Briot, Sébastien Charpentier, Olivier Marin, Pierre Sens (2002) *A Fault-Tolerant Multi-Agent Framework* In: Proceedings of the First International Joint Conference on Autonomous Agents and Multiagent Systems (AAMAS'2002) - Part 2 (Posters) Edited by:Cristiano Castelfranchi, W Lewis Johnson. 672-673 ACM Press.
39. Olivier Marin, Pierre Sens, Jean-Pierre Briot, Zahia Guessoum (2001) *Towards Adaptive Fault-Tolerance For Distributed Multi-Agent Systems* In: 4th European Seminar on Advanced Distributed Systems (ERSADS '2001) 195-201.
40. Zahia Guessoum, Jean-Pierre Briot, Pierre Sens, Olivier Marin (2001) *Toward Fault-Tolerant Multi-Agent Systems* In: Poster Proceedings of the European Workshop on Modeling an Autonomous Agent in a Multi-Agent World (MAAMAW'01) Edited by:Yves Demazeau, Francisco J Garijo. .
41. Damien Cailliau, Arthur Léger, Olivier Marin, Bertil Folliot (2001) *A joint middleware/configuration language approach for space embedded software update* In: Data Systems In Aerospace.

Invited Talks

- Invited talk: "Leveraging distributed quantum computing for sustainable AI", Duke Kunshan University, Kunshan, China, May 2025
- Guest Lecture: "Processes in modern operating systems", Shanghai Jiao Tong University, Shanghai, China, December 2025
- Invited talk: "Leveraging distributed quantum computing for sustainable AI", Duke Kunshan University, Kunshan, China, May 2025
- Invited talk: "Programmable Network Hardware as a Service", Polytechnic Institute of Paris, Palaiseau, France, January 2025
- Guest keynote: "Quantum advantage here & there: distributing the QuBits", International Conference on Information Systems and Computing Technology (ISCTech), Xi'an, China, November 2024
- Invited talk: "Guiding principles toward integrating AI in education", American Chamber of Commerce, Shanghai, China, September 2023

- Guest keynote: "On the Exploitation of the Honest Masses: Trustworthiness as a Scalable Service", International Conference on Computer science, Electronic information engineering, and Intelligent control technology (CEI), Shanghai, China, September 2021
- Invited talk: "GeoTrie: a large scale geospatial-temporal index for massive geotagged data sets", Fudan University, Shanghai, China, June 2017
- Seminar at Microsoft Research Asia, Beijing, China, June 2017
- Invited talk: "Scalability of trust-based systems", East China Normal University, Shanghai, China, April 2017
- Seminar at Microsoft Research Asia, Beijing, China, June 2016
- Seminar at Tsinghua University, Beijing, China, December 2015
- Invited talk: "Building dependability on top of recommendations", Chilean Computer Society, Talca, Chile, November 2014
- Seminar at Yahoo! Labs, Santiago de Chile, Chile, April 2013
- Seminar at the Universidade Federale de Sao Carlos, Sao Carlos, Brazil, March 2012
- Seminar at Universidad Frederico Santa Maria, Valparaiso, Chile, February 2009
- Seminar at Universidad Frederico Santa Maria, Valparaiso, Chile, June 2007
- Invited talk: "DARX - a self-healing framework", 12th Monterey Workshop on Reliable Systems for Unreliable Networked Platforms, Monterey, CA, USA, June 2005
- Invited talk: "Fault tolerance in scalable agent support systems: integrating DARX in the AgentScape framework", Workshop on Agent based Cluster and Grid Computing, Tokyo, Japan, May 2003
- Seminar at the Vrije Universiteit, Amsterdam, The Netherlands, March 2002

Service & Recognition

- University awards: Excellence in Teaching (2010), Excellence in Research (2009, 2013)
- Over 60 Program Committee memberships since 2003 (ICDCS, ICPADS, NCA, etc.)
- Invited talks at Microsoft Research Asia, Tsinghua, Fudan, Yahoo Labs Chile, among others
- Proceedings Chair, Eurosys 2010; Tutorial Chair, SOSP 2017; General Chair, iThings 2026.
- Editorial Board, IEEE Distributed Systems Online (2003–2008)

Education

- **Doctor Europaeus PhD in Computer Science**, Université du Havre, France – 2003
- **MSc in Distributed Systems**, Université Paris 6 - Pierre & Marie Curie, France – 2000

Languages

English (fluent); **French** (native); **Chinese** (basic proficiency; working toward improvement)